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Week 13 Lectures
13 Quiz
Quiz: Week 13 Quiz
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1. Radar remote sensing is made possible principally because (choose the most appropriate one): **1 / 1 point**

- clouds are transparent at microwave frequencies
- there is negligible natural microwave energy radiated from the earth
- there is very little interference with ground-based transmitters for other services

Correct
That is correct, otherwise there would be interference of the natural and radar radiated signals

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2. In synthetic aperture radar (choose one): **1 / 1 point**

- azimuth resolution is a function of slant range to the target
- both ground range resolution and azimuth resolution are both independent of platform height
- because ground range resolution is derived from slant range resolution it is dependent on platform altitude

Correct
That is correct, and is a major advantage of SAR.

3. From an operational perspective an imaging radar for remote sensing purposes is best described by (choose one): **1 / 1 point**

- operating frequency, polarisation configuration, range of incidence angles
- operating frequency, swath width, range of incidence angles
- operating frequency, swath width, altitude

Correct
That is correct. These represent the full range of parameters of interest to the user of SAR data

4. In radar imagery speckle is caused by (choose one): **1 / 1 point**

- the presence of a dominant scatterer within a radar resolution cell
- interference among all the incremental scatterers in a radar resolution cell
- sunlight interfering with the reflected radar energy from the scatterers in a radar resolution cell

Correct
That is correct

5. The scattering coefficient is a property of (choose the most appropriate one): **1 / 1 point**

- the type of radar being used
- a hard target, such as a corner reflector
- earth surface features which scatter radar energy

Correct
That is correct. It is a surface property.

6. Imaging radars based on the principle of synthetic apertures are side looking because (choose one): **1 / 1 point**

- that allows good azimuth resolution to be achieved
- ground range resolution is an inverse function fo the sine of the incidence angle
- better views of ground features are then possible

Correct
That is correct.

7. The polarisation vector of an electromagnetic wave, such as used in imaging radar (choose one): **0 / 1 point**

- points at right angles to the direction of propagation
- points in the direction of propagation
- can be either of the above answer depending on the phase angle of the electric field

Incorrect
That is incorrect. It can only be transverse to the direction of propagation for a wave travelling in free space, such as radar, irrespective of phase angles

8. is microwave energy from the sun possible at the earth's surface? **1 / 1 point**

- no
- yes, at a level much higher than from the earth itself
- yes, at a level much lower than from the earth itself

Correct
That is correct. The microwave power density from the sun is reduced by the inverse square of distance from the sun.

9. The received power density in a radar system is dependent on (choose one): **1 / 1 point**

- the inverse square power of the distance to the target
- the inverse fourth power of the distance to the target
- the inverse sixteenth power of the distance to the target

Correct
That is correct and is a celebrated result from the radar range equation.

10. If the polarisation configuration of a recorded radar image is described as HV, that means (choose one): **1 / 1 point**

- the energy incident on ground features was vertically polarised and the measured energy scattered from the ground had horizontal polarisation
- the energy incident on ground features was horizontally polarised and the measured energy scattered from the ground had vertical polarisation
- both the vertically polarised and horizontally polarised components of the energy scattered from the earth were measured to form the image

Correct
That is the correct interpretation of the subscripts.